INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

ROPE, KERNMANTLE

The General Services Administration has authorized the use of this commercial item description.

1. CLASSIFICATION

1.1 Classification. This commercial item description covers nylon dynamic and static kernmantle ropes in the following types as specified (see 5.2).

Type I - Dynamic kernmantle rope shall be 7 mm in diameter and 150 ± 6 feet (46 ± 2m) long in uninterrupted length. The minimum breaking strength shall be 1984 lbs. (900 kg).

Type II - Dynamic kernmantle rope shall be 8 mm in diameter and 150 ± 6 feet (46 ± 2m) long in uninterrupted length. The minimum breaking strength shall be 2756 lbs. (1250 kg).

Type III - Dynamic kernmantle rope shall be 9 mm in diameter and 150 ± 6 feet (46 ± 2m) long in uninterrupted length. The minimum breaking strength shall be 3472 lbs. (1575 kg). The kernmantle rope shall be approved by the U.I.A.A. (Union Internationale des Associations d'Alpinisme) to hold a minimum of 7 U.I.A.A. (see 3.6) falls and impacts a minimum force of 750 daka Newton (daN).

Type IV - Dynamic kernmantle rope shall be 11 mm in diameter and 165 feet ± 7 feet (50 ± 2m) long in uninterrupted length. The minimum breaking strength shall be 5291 lbs. (2400 kg). The kernmantle rope shall be approved by the U.I.A.A. (Union Internationale des Associations d'Alpinisme) to hold a minimum of 9 U.I.A.A. (see 3.6) falls and impacts a minimum force of 1200 daN on a test weight. The kernmantle rope shall be water repellent treated to resist water penetration into the rope. The maximum average dynamic absorption shall be 26.5 percent.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, ATTN: STRNC-ESC, Natick, MA 01760-5014.

AMSC N/A

FSC 4020

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Type V - Dynamic kernmantle rope shall be 11 mm in diameter and 250 feet ± 10 feet (77 ± 3m) long in uninterrupted length. The minimum breaking strength shall be 5291 lbs. (2400 kg). The kernmantle rope shall be approved by the U.I.A.A. (Union International des Associations d'Alpinisme) to hold a minimum of 9 U.I.A.A. (see 3.6) falls and impacts a minimum force of 1200 daN on a test weight. The kernmantle rope shall be water repellent treated to resist water penetration into the rope. The maximum average dynamic absorption shall be 26.5 percent.

Type VI - Static kernmantle rope shall be 11 mm in diameter and a minimum of 150 feet ± 6 feet (46 ± 2m) long in uninterrupted length. The minimum breaking strength shall be 6500 lbs. (2948 kg.) and the working elongation not greater than 10 percent under a load of 800 lbs. (363 kg). The breaking elongation at rupture shall be greater than 20 percent.

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The kernmantle rope shall be constructed with a nylon core (kern) and a braided outer nylon or polyester sheath (mantle). All ends of the kernmantle ropes shall be cut with a hot knife and taped to prevent fraying. The mantle color shall be Olive Drab No. 7 for the dynamic ropes (types I through V), and black for the static kernmantle rope (type VI).

2.2 Identification. Each kernmantle rope shall have a ticket (identification tag) attached for identification purposes. The ticket shall conform to the requirements for type B, class 1, size 4 or 5, 15 CSU grade designation of UU-T-81. The ticket shall be legibly printed, stamped, or typed with water insoluble ink. The ticket shall contain the following information:

National Stock Number
Nomenclature
Document number
Type
Contract number and date
Name of manufacturer
Date of manufacture (month and year)

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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3.3 End item visual examination. Each lot of kernmantle ropes shall be examined for consumer type defects, defined as follows:

Defects in design, construction, workmanship; Any cut, chafe, damage or splice; kink, darting yarn, knot, broken or loose end, bulged strand; color not as specified; spotty or uneven dyeing; any spot or stain clearly noticeable; identification omitted, incorrect, or not as specified.

The lot size shall be expressed in units of ropes. The sample unit shall be one rope. The inspection level shall be II, and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

3.4 End item dimensional examination. Each lot of ropes shall be measured for diameter and length without damaging or disassembling the end item. Any dimension not within the specified tolerances shall be classified as a defect. The lot size shall be expressed in units of ropes. The sample unit shall be one rope. The inspection level shall be S-3, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

3.5 End item testing. Types I through VI dynamic and static kernmantle ropes shall be tested for minimum breaking strength (see 1.) and elongation in accordance with test method 6016 of Federal Standard No. 191.

3.6 Types III, IV, and V rope impact test. The types III, IV, and V ropes shall be impact tested in accordance with U.I.A.A. Fall Test procedures to determine the impact force on the weight and the number of falls the rope will sustain. An 80 kg. weight of solid iron is dropped from a height of 5 meters. It is held by a section of rope passed through a carabiner with 2.8 meters of rope extending past the carabiner and attached to the weight suspended 2.2 meters above the carabiner. Two measurements are taken, the number of falls held (five minimum), and the impact force on the weight (750 daN minimum for type III, 1200 daN minimum for types IV and V). Failure of the rope to withstand five falls or failure to meet the minimum impact force requirement shall constitute failure of the test. Any industrial testing machine may be used to determine compliance with the requirement.

3.7 Types IV and V rope dynamic absorption test. Types IV and V rope shall be tested for dynamic water absorption in accordance with test method 4500 of Federal Standard No. 191, "Textile Test Methods", except for the following changes to the method:

1. A hot knife (120 Volt Unger #7760 or equivalent, with a 5/16 inch flat tip heating element) shall be used to cut the specimens and seal the rope to prevent unraveling of the ends.

2. In lieu of using a wringer, at the end of the required running time, one specimen shall be grasped at one end and the opposite end tapped downward against a solid object four times. The specimen shall be turned 180°, grasped at the opposite end, and similarly tapped four times at the point previously held. The tapping procedure shall be repeated one additional cycle for each end.

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The specimen shall immediately be placed (all pieces laid side to side) smoothly between two sheets of blotting paper. The specimen and blotting papers shall be firmly hand-pressed for not more than 5 seconds.

The specimen shall be left between the two blotters for a minimum of 3 minutes and a maximum of 4 minutes.

4. PACKAGING

4.1 Packaging. Each length of rope shall be coiled, flattened and tied into hanks not exceeding 24 inches in length. Each coiled length of rope shall then be unit packed in a snug-fitting 200-pound test corrugated fiberboard box or a sealed polyethylene bag with a wall thickness that exceeds 0.003 inches.

4.2 Packing. Four hanks of rope of one type, packaged as specified, shall be packed in a close-fitting RSC style corrugated fiberboard shipping container conforming to type CF, class domestic, grade 275 of PPP-B-636. The shipping container shall be completely filled, without voids, and of a size that can be palletized in accordance with MIL-STD-147. Each shipping container shall be closed in accordance with the appendix of PPP-B-636.

4.3 Palletization. When specified (see 5.2), rope packed as specified shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D, or film bonding F or G, in accordance with MIL-STD-147.

4.4 Marking. In addition to any special markings required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

5. NOTES

5.1 Intended use. The dynamic kernmantle ropes (types I through V) are intended for use in mountaineering operations where a high amount of elongation is desirable. The type VI static kernmantle rope is intended for use in rescue operations and bridging where a low amount of elongation is desirable under a working load.

5.2 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this document.
b. Type required (see 1.).
c. When palletization is required (see 4.3).

5.3 Sources of documents.

5.3.1 Source of government documents. Copies of Military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

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5.3.2 Source for nongovernment document.

Union Internationale des Associations d'Alpinisme (UIAA)
c/o S.A.C. Helvetia Platz-4
Ch.3005 Berne, Switzerland

MILITARY INTERESTS:

Custodians:

Army - GL
Navy - AS

Review activity:

DLA - IS

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - FSS

PREPARING ACTIVITY:

Army - GL

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